

Review

Malodorous Gases in Aquatic Environments: A Comprehensive Review from Microbial Origin to Detection and Removal Techniques

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Abstract: Malodorous gases—particularly hydrogen sulfide (H₂S), ammonia (NH₃), and volatile sulfur compounds (VSCs)—significantly degrade water quality, threaten public health, and disrupt ecosystems. Their production stems from microbial activity, nutrient overload, and industrial discharges, often magnified by low dissolved oxygen. This review integrates current insights into the microbial sulfur and nitrogen cycles to elucidate how these gases form, and surveys advances in detection technologies such as gas chromatography and laser-based sensors. We also assess diverse mitigation methods—including biotechnological approaches (e.g., biofilters, biopercolators), physicochemical treatments, and chemical conversion (Claus Process)—within relevant regulatory contexts in Colombia and worldwide. A case study of the Bogotá River exemplifies how unmanaged effluents and eutrophication perpetuate odor issues, underscoring the need for integrated strategies that reduce pollution at its source, restore ecological balance, and employ targeted interventions. Overall, this review highlights innovative, policy-driven solutions and collaborative efforts as pivotal for safeguarding aquatic environments and surrounding communities from the impacts of odorous emissions.

Keywords: malodorous gases; water quality; microbial processes; detection techniques; removal techniques



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1. Introduction

Water pollution caused by human activities is a global issue that severely impacts water quality, limiting its use for essential applications such as agricultural irrigation, human consumption, and industrial processes such as energy generation. Generally, an estimated 48% of wastewater is released untreated, with higher rates observed in developing regions [1]. In Latin America, outdated wastewater treatment plants and inadequate sanitation infrastructure exacerbate this issue, releasing contaminants of emerging concern into the environment, particularly near urban areas [2]. An illustrative case is the Bogotá River, which traverses both rural and densely populated areas, receiving untreated domestic, industrial, and agricultural wastewater [3]. This contamination not only degrades the ecosystem but also causes the

generation of unpleasant odors due to compounds such as total reduced sulfur (TRS) and ammonia (NH_3), further complicating its management and sustainable use.

Malodorous gases emanating from aquatic environments represent a critical challenge to water quality and public health, necessitating a comprehensive understanding of their sources, detection, and control. Although previous reviews have examined specific aspects of this problem, such as the microbial processes involved in the generation of hydrogen sulfide (H_2S) [4] or methods for the removal of ammonia (NH_3) from wastewater [5], a holistic approach integrating the full spectrum of malodorous gases, including volatile sulfur compounds (VSCs), has been lacking. In addition, recent advances in detection technologies, particularly in sensor development and spectroscopic techniques, together with innovative biotechnological and chemical mitigation strategies, warrant a timely and comprehensive update of the existing literature. This review addresses this gap by providing an interdisciplinary synthesis of the current knowledge on malodorous gases in aquatic systems, including their microbial origins, advanced detection methods, various removal techniques, and relevant regulatory frameworks. In addition, we incorporate a case study of the Bogotá River to illustrate the real-world complexities and practical implications of odor pollution, a perspective often absent in previous reviews. This integrated approach offers valuable information for researchers, policymakers, and practitioners striving to mitigate the environmental and social impacts of malodorous gas emissions from aquatic environments.

Previous research has addressed the issue of foul-smelling gases in aquatic environments, but typically in a fragmented manner. For example, reviews have been published focusing exclusively on hydrogen sulfide (H_2S) generated by microbial processes in sewage systems [6] or assessments of methods for hydrogen sulfide removal [7]. There are also specific studies on volatile sulfur compounds (VSCs) in inland waters—for instance, ref. [8] examined the sources and sinks of foul-smelling sulfur compounds in lakes and rivers.

However, an integrated perspective encompassing all these odorous gases together was lacking. Until now, no holistic review had integrated both sulfur compounds (H_2S and VSCs) and nitrogen compounds (NH_3), along with their microbial mechanisms, advanced detection methods, and control strategies. This review aims precisely to fill that gap.

Notably, this work provides an interdisciplinary synthesis of the current knowledge, covering everything from the microbial origins of these gases to the latest monitoring technologies and relevant regulatory initiatives. Additionally, it incorporates a real-world case study (the Bogotá River), illustrating the complexities and challenges of odors in a contaminated water body—something rarely addressed in previous reviews. Thus, the introduction establishes that this review will offer an integrated and novel perspective, valuable for both researchers and environmental managers, filling knowledge gaps and guiding solutions.

Although ammonia (NH_3) and other volatile sulfur compounds (VSCs) also contribute to malodors in aquatic environments, this review focuses primarily on hydrogen sulfide (H_2S). The rationale for this scope is twofold: First, H_2S presents a more immediate toxicity risk at relatively low concentrations, often making it the highest-priority gas to mitigate in industrial and municipal contexts. Second, many NH_3 -focused remediation methods are substantially more complex or cost-intensive, and thus beyond the current scope. While we acknowledge the significance of NH_3 and VSC removal strategies—especially in areas where these gases dominate malodor concerns—detailed analyses of their treatment mechanisms, operational costs, and technological challenges warrant a separate in-depth review. Readers interested in advanced NH_3 or VSC control methods may consult the existing literature [9,10] for comprehensive assessments of associated technologies and cost-effectiveness.

2. Emission Sources of Malodorous Gases

2.1. Total Reduced Sulfur (TRS) and Ammonia (NH_3) as Major Malodorants

TRS compounds consist of a group of sulfur-containing gases that are primarily known for their intense odor and low olfactory thresholds [10]. These compounds, which include hydrogen sulfide (H_2S), mercaptans (e.g., methyl mercaptan (CH_3SH) and dimethyl disulfide (DMDS)), are typically associated with anaerobic decomposition in organic-rich environments [8].

TRS compounds are a subset of volatile sulfur compounds (VSCs), a broader class of sulfur gases emitted during microbial activity and chemical reactions in wastewater treatment, landfills, and industrial facilities [11]. Among these, H_2S is the most studied TRS compound, mainly due to its distinctive “rotten egg” odor with an extremely low odor threshold ($11.1 \mu\text{g}/\text{m}^3$) [12]. However, other TRS compounds, such as mercaptans and DMDS, also present considerable challenges. Methyl mercaptan (CH_3SH) has an odor threshold ranging from 0.02 to 0.4 ppbv [13], which is perceived as more pungent than H_2S and contributes significantly to complaints of malodor [14].

In addition to TRS, NH_3 is a highly volatile and reactive gas that plays a significant role as an offensive odorant in various environments, particularly in agricultural, industrial and waste management activities [15]. NH_3 is characterized by its sharp and pungent smell and can be detected at concentrations as low as 5 ppm [16]. Its volatility and widespread sources contribute significantly to complaints about malodors and environmental degradation, which requires effective monitoring and mitigation measures.

NH_3 has significant environmental consequences, mainly due to its role as a precursor to fine particulate matter ($\text{PM}_{2.5}$). Once released into the atmosphere, NH_3 reacts with acidic compounds, including sulfuric acid (H_2SO_4) and nitric acid (HNO_3), to form ammonium salts, which are a significant component of ($\text{PM}_{2.5}$) [17]. These fine particles contribute to air pollution, reduce visibility, and pose significant health risks. Furthermore, the deposition of NH_3 and its reaction products in soils and water bodies disrupt the balance of nutrients, promoting eutrophication and acidification in sensitive ecosystems [18]. This process disrupts aquatic habitats and reduces biodiversity, illustrating the dual role of NH_3 as a local and regional environmental pollutant.

2.2. Emission Sources

2.2.1. Anthropogenic Emissions

One of the primary sources of H_2S emissions is the oil and natural gas industry, through processes such as exploration, extraction, refining, and transportation [19]. In each of these processes, H_2S can be released as a by-product, particularly during petroleum refining and desulfurization, and in the processing of natural gas, where chemical and thermal separations are employed [20]. Another significant source of emissions is wastewater treatment, where activated sludge is generated to promote the degradation of organic matter by microorganisms [21]. During this process, H_2S is produced as a by-product of microbial anaerobic decomposition and is one of the primary compounds responsible for odors.

In sewer systems and wastewater treatment plants, the accumulation of organic matter under anaerobic conditions generates H_2S and VSCs in high concentrations. Field studies in treatment plants (WWTPs) report that in confined spaces (e.g., well heads, digesters), H_2S levels can reach hundreds of ppm [22]. For example, in a municipal plant, H_2S concentrations in the inlet air of a biofilter were measured between 200 and $1300 \text{ mg}/\text{m}^3$, approximately equivalent to 140 to 930 ppm [22].

Agriculture is the predominant source of global NH_3 emissions, accounting for approximately 80% of anthropogenic releases [23]. The application of fertilizers is a key source of NH_3 emissions [24]. Under certain environmental conditions, particularly high

temperatures and alkaline soils, ammonium-based fertilizers undergo volatilization, releasing NH_3 into the atmosphere [25]. In industrial contexts, NH_3 is emitted from chemical manufacturing and wastewater treatment processes [26]. Furthermore, the combustion of fossil fuels and biomass also releases NH_3 , contributing to its presence in urban and peri-urban areas [27]. These sources highlight the widespread prevalence of NH_3 emissions in various sectors.

In industrial settings, NH_3 is released to a lesser extent, for instance, in chemical manufacturing processes or fertilizer storage. However, in intensive livestock facilities, concentrations can be very high: measurements inside pig or poultry barns typically report between 5 and 18 ppm of NH_3 , with peaks reaching up to 47 ppm under poor ventilation conditions [28].

2.2.2. Natural Emissions

Natural sources of H_2S emissions include geothermal sources such as geysers and hot springs, which release H_2S as a by-product of volcanic and geothermal activity [29]. Volcanoes, in particular, release H_2S during eruptions, contributing to the accumulation of this gas in the atmosphere [30]. Another source is the anaerobic decomposition of organic matter by microorganisms, which also generates H_2S in natural environments such as swamps, peat bogs, and standing bodies of water [31]. In these ecosystems, microorganisms break down organic matter without oxygen, producing H_2S as one of the resulting compounds.

For example, in rivers highly eutrophicated by untreated wastewater, the formation of anoxic zones in sediments is common, continuously releasing H_2S and ammonia. A study by the ATSDR agency reports that in polluted watercourses, dissolved ammonia concentrations of up to 16 ppm (mg/L) have been detected [28], compared to just 0.01–0.043 ppm in unaffected surface waters. These high levels occur near industrial or urban effluent discharge points, contributing to urine- or ammonia-like odors along the shorelines.

Natural sources of NH_3 emissions include the decomposition of organic matter and biological processes in the environment [32]. A significant source is the microbial breakdown of nitrogen-containing compounds, such as urea and proteins, in soils and water bodies [33]. Wetlands, marshes, and other aquatic ecosystems contribute to NH_3 emissions due to the anaerobic degradation of organic material [34]. Furthermore, the excretion of urea by animals, particularly in natural habitats or unmanaged grazing systems, leads to its hydrolysis by urease enzymes, releasing NH_3 into the atmosphere [35].

In natural aquatic ecosystems (wetlands, swamps, estuaries), the generation of foul-smelling gases primarily originates from geothermal processes or the anaerobic decomposition of organic matter. Typical background levels of H_2S in natural ambient air are very low (on the order of 0.00011 to 0.00033 ppm, i.e., 0.11–0.33 ppb) [36]. However, in locations with volcanic or geothermal activity, these values can increase significantly. For example, in areas with hot springs and fumaroles, H_2S concentrations of up to several ppm have been measured; a study in Rotorua (New Zealand) recorded 7–8 ppm of H_2S near the ground in geothermal emission areas [37].

In stagnant natural waters with a high organic load, the lack of dissolved oxygen leads to the biogenic production of H_2S by sulfate-reducing bacteria. Volatile sulfur compounds (VSCs) resulting from microbial degradation (e.g., methanethiol, dimethyl sulfide, dimethyl disulfide) are usually detected in trace amounts, but they are sufficient to cause odors due to their extremely low olfactory thresholds [38]. In fact, the proliferation and subsequent decay of algae in freshwater bodies can release large quantities of foul-smelling VSCs—including H_2S , mercaptans, and organic sulfides—during the anaerobic putrefaction of biomass [39]. A documented case involves episodes of “black water” in eutrophic lakes, where the decomposition of cyanobacterial blooms generates odors reminiscent of rotten eggs and decomposing organic matter.

2.2.3. Hybrid Emissions

Some industrial processes, by their nature, facilitate the emission of H₂S generated naturally but enhanced by human activity. One such example is the use of wastewater in industrial processes, where the movement and treatment of this water lead to the release of H₂S into the environment [40], particularly in still tanks in water treatment or power generation plants. The turbulence and agitation of water during its movement cause the release of H₂S into the atmosphere [41]. Another example is anaerobic digestion in biogas plants, where organic waste decomposes without oxygen, generating H₂S as one of the by-products [42].

The storage and spread of manure and slurry, particularly in open systems, promote the microbial degradation of nitrogen compounds, resulting in significant NH₃ releases [43]. In wastewater treatment plants, the breakdown of nitrogenous waste during biological treatment can also enhance NH₃ emissions, especially during stages of aeration or agitation [44]. Furthermore, composting organic waste under aerobic and anaerobic conditions releases NH₃, combining natural microbial processes with industrial waste management practices [45]. These hybrid sources underscore the role of human activity in amplifying natural NH₃ emissions.

Table 1 presents a comprehensive comparison between the emissions sources.

Table 1. Comparison between emission sources.

Category	Examples of Sources	Main Emitted Gases	Typical Emission Levels	Impact on Air Quality
Natural	Swamps, wetlands, thermal springs, volcanic activity, anaerobic decomposition	H ₂ S, NH ₃ , VSC (dimethyl sulfide, methyl mercaptan)	0.00011–0.00033 ppm (H ₂ S in wetland air); up to 7–8 ppm near geothermal sources	Low to moderate; depends on environmental conditions
Anthropogenic	Wastewater treatment plants, agriculture (livestock, fertilizers), chemical industry, oil refineries	H ₂ S, NH ₃ , VSC, volatile organic compounds (VOCs)	5–18 ppm (NH ₃ in pig farms); 200–1300 mg/m ³ (H ₂ S in wastewater treatment plants)	High; contributes to strong odors in urban and rural areas
Hybrid	Rivers polluted by urban and industrial discharges, waste treatment in landfills, biogas production	H ₂ S, NH ₃ , VSC, gas mixtures depending on contamination	Up to 16 ppm NH ₃ in polluted rivers; VSC in tens of ppb in landfills	Variable; depends on pollution levels and proximity to emission sources

3. Regulatory Framework and Odor Thresholds

3.1. International Standards and Legislation

Numerous countries have established specific regulations and guidelines to monitor and control the emission of odorous compounds. In Europe, the UNE-EN 13725:2022 standard [46], which reviews the 2004 iteration, outlines methodologies to determine malodor concentration through dynamic olfactometry, serving as a standard reference framework for assessing odor emissions throughout the European Union. This standard has been extensively adopted by countries outside Europe, including Australia, Chile, Colombia, the United States, South Korea, and New Zealand, thus improving the consistency and comparability of measurements worldwide.

In addition to UNE-EN 13725, additional standards, including UNE-EN 15259 [47] and ISO 16911-1 [48], provide methodologies for organizations and laboratories involved in the

quantification and regulation of odorous emissions. These standards delineate protocols for sampling, measurement, and data interpretation, facilitating a more precise assessment of the impact of malodor on adjacent communities.

In Asia, nations such as Australia [49], China, and Japan also have specific regional and sectoral regulations. For example, Japan's Offensive Odor Control Law, amended in 1995, aims to enhance effectiveness. Several other countries, including Canada (Ontario [50] and Alberta [51]), the US, and Argentina's Buenos Aires province, have established regional emission standards tailored to specific activities in their jurisdictions.

In Latin America, Chile launched a sectoral Odor Management Strategy in 2012 under its environmental regulation program [52]. By 2023, regulations for key sectors such as pork and fishing were published and adopted [53].

According to the review, most legislative initiatives do not establish specific maximum thresholds for point activities or facilities with potential odorous impact. Instead, the regulations focus on monitoring and tracking methodologies, using olfactometric studies, dispersion modeling or other assessment methods. As a result, reference thresholds are often expressed in odor units (EUo) rather than defined concentration limits for a specific odor-generating substance.

3.2. Regulation and Air Quality Standards in Colombia

In Colombia, Law 9 of 1979 sets general guidelines for preserving and improving sanitary conditions, environmental quality, and public health [54]. Later, in 1995, Decree 948 established regulations to protect and control air quality [55]. However, it was not until Resolution 601 of 2006 that air quality and emission levels were officially regulated, establishing thresholds for key substances that generate offensive odors [56]. Odor is considered offensive when unpleasant and can cause discomfort, although it does not necessarily harm the health.

Resolution 601 was amended by Resolution 610 in 2010 [57], and in 2013, Resolution 1541 expanded the regulations to include the maximum allowed levels and guidelines for evaluating activities that generate offensive odors [58]. These resolutions also adopted European olfactometry standards, which measure odor concentrations and their impact on populations.

In Colombia, the main substances of offensive odor are H_2S , TRS, and NH_3 , with H_2S being the most common in activities that involve the discharge of water, such as hydroelectric generation. Resolution 2087 of 2014 established protocols for monitoring, controlling, and surveilling offensive odors, detailing procedures to assess air quality and emission levels, and using dispersion models for atmospheric pollutants [59]. The NTC 6012-1:2013 standard [60] provides methods to evaluate odors, focusing on the psychometric evaluation of odor annoyance.

4. Thresholds, Exposure Times, and Concentrations of H_2S

Odor thresholds are established based on perception, duration of exposure, and concentration levels to avoid acute and chronic effects. Table 2 summarizes the thresholds and maximum permissible levels.

Table 2. Comparative summary of odor thresholds and safety of H₂S in different countries based in their legislation.

Reference	Application	Exposure Time/Concentration
Odor perception threshold in Toxicological profile AFTS [31]	International reference	0.0005 to 0.13 ppm
WHO guideline for tolerable concentration in air	International reference	150 µg/m ³ (108 ppm)
Odor perception threshold in Spain NTP 320 [61]	Spain	0.0081 ppm
TA Luft Technical Instruction on Air Quality Control	Germany	8 µg/m ³ (0.006 ppm)
Air Quality Guide	New Zealand	7 µg/m ³ (0.005 ppm)
Maximum permissible level in Resolution 610 2010 [57]	Colombia	7 µg/m ³ (0.005 ppm)
Maximum permissible level in Resolution 1541 de 2013 [58]	Colombia	1 h average exposure/30 µg/m ³ (0.021 ppm) 24 h average exposure/7 µg/m ³ (0.005 ppm)

5. Health Effects of Odorous Gases

5.1. Health Effects of TRS

The health impacts of TRS compounds depend on their concentration and duration of exposure. At concentrations exceeding 150 ppm, CH₃SH can cause depression of the central nervous system, nausea, and respiratory irritation [62]. At concentrations as low as 50 ng/mL, CH₃SH inhibits the growth and proliferation of epithelial cells, although it does not affect fibroblasts [63].

DMDS is another major TRS component, recognized for its garlic-like odor and its role in industrial malodor events [64]. Despite its higher odor threshold compared to H₂S and mercaptans, DMDS has the potential to cause acute respiratory irritation and headaches at moderate concentrations [65]. Prolonged exposure can lead to delayed toxicity, including mucosal inflammation in the eyes, nose, oropharynx, and airways, and chronic exposure to lower levels may cause long-term respiratory effects [66].

Chronic exposure to low levels, often found near industrial facilities, can cause symptoms such as fatigue, headaches, and respiratory discomfort [67]. Acute exposure to high concentrations of CH₃SH and DMDS can cause severe respiratory distress, effects on the central nervous system, and, in extreme cases, death [65].

A comparative analysis of the intensity and toxicity of the odor of TRS components shows that each compound contributes a distinctive profile to the overall impact of TRS emissions [68]. H₂S is of particular concern due to its high toxicity and detectability, making it a primary target for mitigation efforts. Although mercaptans are less hazardous to health than H₂S, they are much more potent odorants and contribute disproportionately to complaints of malodor due to their very low detection thresholds [12]. Despite its relatively low impact on the intensity and toxicity of the odor, DMDS has been shown to improve the persistence and complexity of malodors in industrial environments [68]. These differences require the implementation of comprehensive monitoring systems capable of detecting and quantifying each component of the TRS to inform effective control strategies.

5.2. Health Effects of NH₃

The health impacts of NH₃ exposure depend on its concentration and duration. At low concentrations, NH₃ acts as an irritant, causing discomfort in the eyes, nose, and throat [69].

Chronic exposure to NH_3 can have significant health effects. In laying hens, long-term exposure to 30 ppm NH_3 increased plasma levels of certain inflammatory markers [70]. Moreover, NH_3 contributes to the generation of ($\text{PM}_{2.5}$), exacerbating pre-existing cardiovascular and pulmonary conditions, increasing morbidity and mortality in affected populations [71].

6. Biogeochemical and Microbial Processes Leading to Gas Formation

6.1. Biological Sulfur Cycle and Microbial Processes Leading to H_2S Production

The sulfur cycle is an essential biogeochemical process that ensures the availability of sulfur in suitable chemical forms for living organisms and regulates fundamental processes in diverse ecosystems [72]. This cycle (Figure 1) describes the transition of sulfur between various oxidation states. The process is primarily driven by microorganisms that facilitate sulfur cycling in different environments, including extreme habitats, providing opportunities to mitigate sulfur-based pollution [73]. These microorganisms play an essential role in the conversion of sulfur compounds to reduced and oxidized forms through complex metabolic processes [74]. In addition to its role in maintaining environmental balance, the sulfur cycle is vital to biological productivity and has significant applications in wastewater treatment and sulfur pollutant mitigation [73].

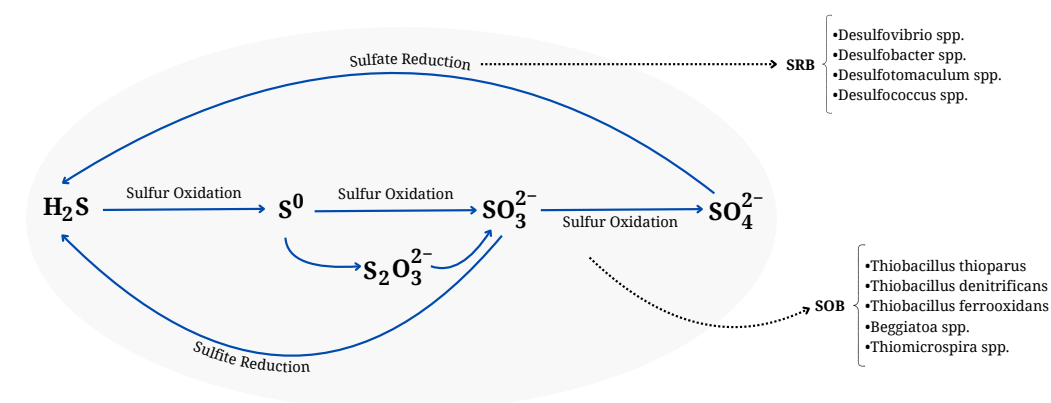
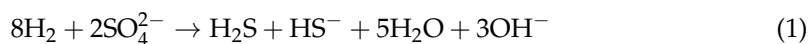


Figure 1. Simplified schematic representation of the pathways occurring in the microbial sulfur cycle. The diagram clearly illustrates key biochemical transformations within the sulfur cycle mediated by sulfate-reducing bacteria (SRB) and sulfur-oxidizing bacteria (SOB). Different oxidation states of sulfur—including hydrogen sulfide (H_2S), elemental sulfur (S^0), thiosulfate ion ($\text{S}_2\text{O}_3^{2-}$), sulfite ion (SO_3^{2-}), and sulfate ion (SO_4^{2-})—are depicted, showing their interconnected transformations through oxidation and reduction reactions. Adapted from [75,76].

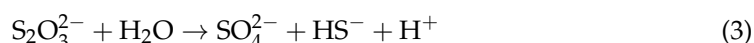
A key component of the sulfur cycle is sulfate (SO_4^{2-}) reduction, carried out by sulfate-reducing bacteria (SRB) under strictly anaerobic conditions. These bacteria are essential for sulfur recycling in sedimentary environments and closed aquatic systems, contributing significantly to sulfur flux within ecosystems [77]. SRB produce H_2S by reducing sulfur compounds, thus facilitating the decomposition of organic matter [78]. For example, in wastewater, H_2S is often used as an indicator of anaerobic degradation of organic matter, helping to identify decomposition processes without oxygen [79]. However, some SRB have been found to respire oxygen, although no studies have shown that pure cultures of SRB thrive under aerobic conditions [80]. In addition, in oxygenated environments, H_2S can

form through sulfurized amino acid degradation [78], highlighting the diversity of microbial mechanisms that generate this compound and its ecological relevance in various ecosystems.

SRB utilize oxidized compounds such as SO_4^{2-} , sulfite (SO_3^{2-}), and elemental sulfur (S^0) as electron acceptors. During this process, SO_4^{2-} is reduced to H_2S while oxidizing organic substrates or hydrogen, which provides energy for cell growth. The general reaction for SO_4^{2-} reduction is as follows.



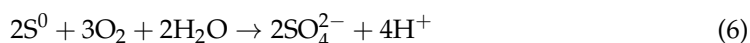
Sulfite reduction is one of the most studied reactions. In environments suitable for SRB, sulfite reduction and thiosulfate reduction ($\text{S}_2\text{O}_3^{2-}$) can also occur [81].



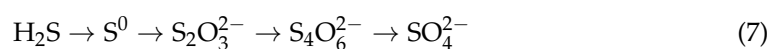
In contrast, sulfur-oxidizing bacteria (SOB) complement the activity of SRB by transforming reduced sulfur compounds into oxidized forms. These autotrophic bacteria convert H_2S into elemental sulfur (S^0) and, eventually, to SO_4^{2-} [82]. This oxidative process can occur under aerobic conditions or limited oxygen availability, as described by the following reactions.



Elemental sulfur (S^0) can also be oxidized to SO_4^{2-} through metabolic processes carried out by specialized microorganisms. The oxidation of S^0 is primarily mediated by chemolithotrophic and photoautotrophic microorganisms [83], which use molecular oxygen as the final electron acceptor.



During the oxidation of H_2S to SO_4^{2-} , SOB generate a series of intermediate compounds that are crucial for the dynamics of the sulfur cycle. This stepwise oxidation process, characteristic of genera such as *Thiobacillus*, includes the formation of $\text{S}_2\text{O}_3^{2-}$ and tetrathionate ($\text{S}_4\text{O}_6^{2-}$) as intermediates before reaching the final form SO_4^{2-} . The general sequence of reactions is represented as follows.



In the first stage, H_2S is oxidized to elemental sulfur (S^0), which can accumulate temporarily inside or outside the cells, depending on the microbial species. For example, species such as *Thiobacillus* and *Chlorobiaceae* deposit S^0 outside the cell [84], while others, such as *Beggiatoa*, *Thiothrix*, and *Chromatiaceae*, store it intracellularly [85]. Afterward, S^0 is converted into $\text{S}_2\text{O}_3^{2-}$, which is then transformed into $\text{S}_4\text{O}_6^{2-}$ before being fully oxidized to SO_4^{2-} . Each of these steps is catalyzed by specific enzymes that optimize electron transfer and energy flow under varying environmental conditions [76].

In environments where oxygen availability is limited, SOB exhibit metabolic versatility using alternative electron acceptors, such as nitrate (NO_3^-), during H_2S oxidation [80]. This mechanism enables anoxic sulfide oxidation, particularly in environments such as biofilms, stratified sediments, and eutrophic aquatic systems with steep oxygen gradients.

Sulfur also exists in organic forms within biomolecules such as amino acids, integrating into the sulfur cycle through assimilation and decomposition processes. During organic

matter decomposition, sulfur is released as H_2S or volatile compounds such as dimethyl sulfide (DMS), contributing to the sulfur flux in the atmosphere [86].

Finally, in ecosystems such as marine sediments and sewage biofilms, interactions between SRB and SOB create a tightly coupled cycle. The sulfide produced by SRB diffuses into oxygenated zones, where SOB oxidize it back to SO_4^{2-} , ensuring a continuous flow of sulfur. This mechanism, essential in both natural and engineered systems, is the foundation of wastewater treatment and environmental remediation technologies.

6.2. Formation Mechanisms of TRS Compounds

TRSs are produced by biochemical and microbiological processes that involve the activity of heterotrophic and anaerobic microorganisms, as well as chemical reactions catalyzed under specific conditions. Although H_2S production is typically associated with anoxic environments, recent research has shown that certain freshwater bacteria, such as the *Stenotrophomonas* and *Chryseobacterium* species, can generate H_2S under aerobic conditions by degradation of L-cysteine [78]. In *Escherichia coli*, this process occurs via two discrete metabolic pathways. One pathway involves L-cysteine desulfhydrases, which directly catalyze the production of H_2S . The alternative path involves a series of enzymatic reactions in which L-cysteine is initially transformed by L-cysteine aminotransferase and subsequently by 3-mercapto pyruvate sulfurtransferase, leading to the generation of reactive sulfur species [87]. These pathways illustrate the metabolic flexibility of *E. coli* in the processing of sulfur amino acids, producing volatile compounds such as H_2S or sulfur intermediates involved in redox processes. This demonstrates the diverse mechanisms of H_2S production.

VSCs primarily originate from the degradation of organic sulfur compounds, particularly the amino acids cysteine and methionine, which are released during the decomposition of organic matter. Several studies have identified these amino acids as key precursors for the formation of VSCs, including CH_3SH , DMS, and even H_2S [38]. Although reduction of SO_4^{2-} contributes to the formation of VSCs, the decomposition of organic matter is the primary source in specific environments, such as peatlands [88].

The degradation of cysteine by microorganisms occurs under aerobic and anaerobic conditions, resulting in the release of H_2S , NH_3 , and other compounds. In anaerobic environments, this reaction is catalyzed in archaea like *Methanocaldococcus jannaschii* by cysteine desulfurase, which contains a [4Fe-4S] cluster critical for its catalytic activity [89]. In bacteria such as *Salmonella enterica*, the enzyme CyuA plays a similar role in cysteine degradation under reducing conditions [90]. In addition, methionine is metabolized by methionine gamma-lyase, which converts it to CH_3SH , producing NH_3 and 2-oxobutyric acid [91]. These initial processes not only result in the formation of volatile compounds such as H_2S and CH_3SH but also set the stage for subsequent reactions that lead to other complex VSCs that contribute significantly to the formation of odors in organic-rich environments [88].

In addition to being a direct product of methionine degradation, CH_3SH can also be formed by the methylation of H_2S in the presence of methyl donors such as S-adenosylmethionine [92]. CH_3SH not only serves as an intermediate in VSC synthesis but also has a distinctive odor [93]. In anaerobic systems, CH_3SH can undergo further methylation to form DMS or be oxidized to DMDS [94]. This reaction, mediated by microorganisms adapted to reduce conditions such as methanogens and SRB, involves the combination of two CH_3SH molecules with the release of water [95].

In environments with higher oxidative activity or where anoxic conditions alternate with oxygenic conditions, disulfides can evolve into dialkyl polysulfides [96,97]. These compounds, which contain sulfur atom chains, are formed through the oxidation of disulfides [98]. Polysulfide accumulation is particularly evident in aquatic sediments, where

microbial interactions and chemical reactions favor their production [99]. Although poly-sulfides are less volatile than other VSCs, their presence in sediments can influence the dynamics of sulfur and the generation of other malodorous compounds when environmental conditions change [100].

The formation of dimethylsulfide from dimethylsulfoniopropionate (DMSP), a compound mainly produced by algae and phytoplankton [101], has been identified in aquatic systems [102]. This precursor is metabolized by bacteria through specific enzymatic pathways, such as DMSP lyase, which releases DMS and acrylate [103]. Although this route is more prevalent in marine environments, its contribution in freshwater systems with high salinity or eutrophication should not be overlooked [104]. The metabolic diversity of the microorganisms involved in these processes reflects the complexity of VSC formation mechanisms in different environmental contexts.

The formation of TRS, particularly VSCs, reflects intricate biochemical and microbiological processes that occur in both natural and controlled systems. The interaction between heterotrophic, sulfate-reducing, and methanogenic microorganisms with organic matter and specific environmental conditions leads to a wide diversity of sulfur compounds, each with unique properties and effects. These compounds present significant challenges in wastewater and sediment management while also playing a crucial role in the chemical and biological dynamics of ecosystems with direct implications for air and water quality.

6.3. Biological Nitrogen Cycle and Microbial Processes Leading to NH_3 Production

Like the sulfur cycle, the biological nitrogen cycle (Figure 2) plays a vital role in maintaining ecological balance by facilitating the circulation of essential chemical compounds between different environmental compartments. Similarly to the sulfur cycle, which was discussed earlier, the nitrogen cycle is driven by an intricate network of geochemical processes, redox reactions, and microbial pathways [33].

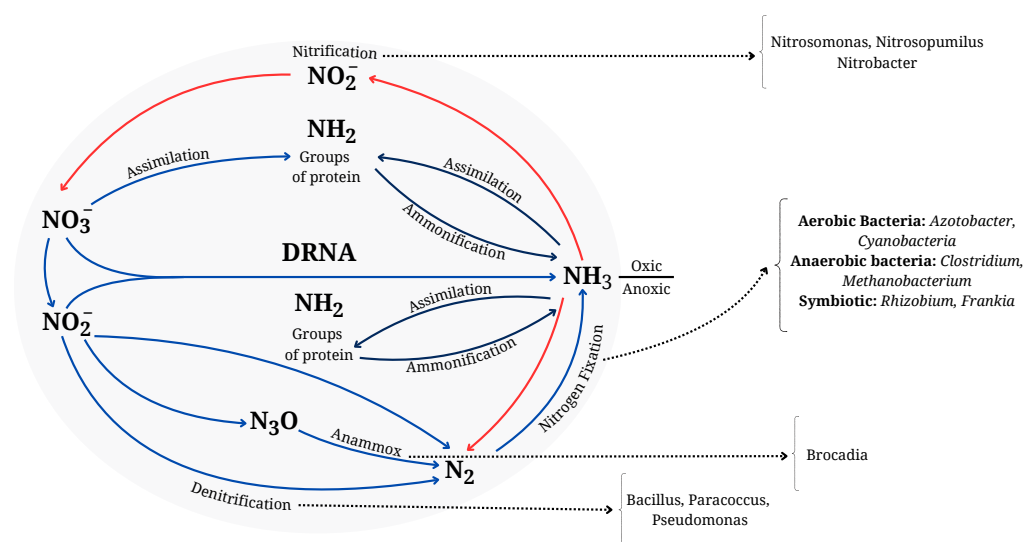
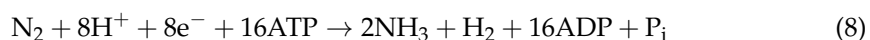


Figure 2. Simplified schematic representation of the microbial nitrogen cycle pathways. Blue arrows indicate reductive processes, and red arrows denote oxidative processes. Key nitrogen species depicted include atmospheric nitrogen (N_2), ammonia (NH_3), organic nitrogen (NH_2 groups within proteins), nitrite (NO_2^-), nitrate (NO_3^-), and nitrous oxide (N_2O). Representative microorganisms involved in specific transformations are provided, highlighting nitrogen fixation (e.g., *Azotobacter*, *Rhizobium*), nitrification (e.g., *Nitrosomonas*, *Nitrobacter*), denitrification (e.g., *Pseudomonas*), and anaerobic ammonium oxidation (*Anammox*, e.g., *Brocadia*). The diagram distinguishes between transformations occurring under oxic and anoxic conditions. Adapted from [105].

One of the most important processes in the nitrogen cycle is biological nitrogen fixation, a reduction reaction in which molecular nitrogen (N_2) from the atmosphere is transformed into NH_3 , a form of nitrogen accessible to living organisms. The fundamental chemical reaction, catalyzed by the nitrogenase enzyme, is as follows.



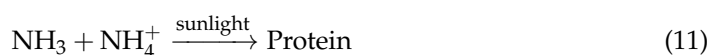
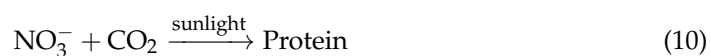
Nitrogenase, a highly conserved enzyme throughout evolution [106], requires considerable energy in the form of ATP and electron transfer to break the triple bond of molecular nitrogen [107]. This process is carried out by a variety of microorganisms, including both bacteria and archaea, in different environments [106,108]. Nitrogen-fixing organisms include aerobes and anaerobes, such as *Azotobacter* and *Clostridium*. In addition, there are widespread symbiotic associations between plant species, classified into three main types: cyanobacterial, actinorhizal, and rhizobial [109]. Examples include associations of *Rhizobium* and *Bradyrhizobium* with legumes and *Frankia* with non-legume plants [110]. These microorganisms develop specialized mechanisms, such as heterocysts or anaerobic microenvironments, to protect nitrogenase from oxygen inhibition [111].

Ammonification is the process by which organic nitrogen in compounds such as proteins, nucleic acids, and urea is converted into NH_3 or its ionized form, ammonium (NH_4^+) [112]. This process is carried out by decomposer microorganisms that degrade organic matter from plants, animals, and animal waste. The general reaction can be represented as follows.



Bacillus subtilis and *Pseudomonas fluorescens* have been shown to increase urease activity in soils, enhancing bioaccessible nitrogen and ammonium content in these environments [113]. This process not only recycles nitrogen into forms that microorganisms and plants can assimilate, but also connects organic matter decomposition to other cycle processes such as nitrification and denitrification [114].

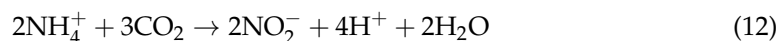
Nitrogen assimilation is a critical step in transforming inorganic nitrogen, such as NH_4^+ and NO_3^- , into essential organic compounds [115]. This process is vital for synthesizing amino acids, proteins, and other nitrogenous compounds required for the growth and function of living organisms [116]. Plants can absorb nitrogen in the form of both NO_3^- and NH_4^+ , although the metabolic pathways for assimilating these forms differ. NO_3^- must first be reduced to nitrite (NO_2^-) and then to NH_4^+ before incorporation into organic molecules [117].



The primary nitrogen incorporation pathway in plants and microorganisms is the glutamine synthetase–glutamate synthase (GS/GOGAT) pathway, a highly conserved mechanism that converts NH_4^+ into organic compounds [115]. In this pathway, NH_4^+ combines with glutamate to form glutamine, which serves as an amino group donor to synthesize other amino acids [118]. This system efficiently incorporates nitrogen into cellular metabolism and regulates nitrogen homeostasis in response to environmental factors such as nutrient availability and light [119].

Nitrification is the microbial conversion of NH_4^+ to NO_2^- and then to NO_3^- through oxidation reactions [107]. This process creates highly mobile plant-accessible forms of nitrogen, carried out exclusively by chemoautotrophic microorganisms in two main stages [120]. First, bacteria such as *Nitrosomonas* and *Nitrosospira* oxidize NH_4^+ to NO_2^- . In the second stage, microorganisms like *Nitrobacter* and *Nitrospira* convert NO_2^- to NO_3^- , a stable com-

pound in aerobic environments [121]. Recent studies have shown that bacteria *Nitrospira* can complete nitrification, oxidizing NH_3 directly to NO_3^- [122]. Nitrification plays a key role in agricultural soils and composting systems [123]. However, it can also cause nitrogen losses due to leaching and gaseous emissions, reducing plant nitrogen availability and contributing to environmental issues such as eutrophication and greenhouse gas production [124].



Denitrification is the process by which NO_3^- is reduced to gaseous forms of nitrogen, including NO_2 , nitric oxide (NO), nitrous oxide (N_2O), and ultimately molecular nitrogen (N_2), which is released into the atmosphere [107]. This process is catalyzed by specialized enzymes, including nitrate reductase, nitrite reductase, nitric oxide reductase and nitrous oxide reductase, whose activity is up-regulated under anaerobic conditions [125]. Denitrification occurs primarily in water-saturated soils, sediments, and wastewater treatment systems, where the absence of oxygen favors the use of nitrates as electron acceptors during cellular respiration [126].

This process is dominated by denitrifying microorganisms, including families such as *Comamonadaceae* and genera such as *Diaphorobacter*, *Acidovorax*, and *Simplicispira*, which are particularly abundant in anaerobic environments [127]. Denitrifiers reduce nitrates and nitrites, closing the nitrogen cycle and playing a vital role in mitigating reactive nitrogen compounds in aquatic and terrestrial environments [128]. However, the release of N_2O , a potent greenhouse gas, during intermediate stages can adversely impact global climate if the process does not proceed to completion to N_2 [129].

Anaerobic ammonium oxidation (anammox) is a microbial process in which bacteria oxidize NH_4^+ using NO_2^- as an electron acceptor, producing N_2 as the final product [130]. The following reaction represents this process.



Anammox bacteria, members of the *Planctomycetes* phylum, are essential for nitrogen loss in various anoxic environments, including marine sediments, wastewater treatment facilities, and wetlands [130]. The process involves intermediates such as nitric oxide (NO) and hydrazine (N_2H_4), with c-type heme proteins facilitating catalytic reactions [131]. A distinguishing feature of anammox bacteria is the anammoxosome, a specialized organelle enclosed by ladderane lipids. This organelle contains hydrazine oxidoreductase, an enzyme critical for the conversion of hydrazine to dinitrogen gas [132].

Anammox bacteria contribute significantly to nitrogen loss in oceanic systems, influencing global nitrogen cycles. Their application in engineered systems, particularly in wastewater treatment, has gained attention due to their efficiency in removing NH_3 from anaerobic systems [132]. Anammox activity can be established in anaerobic membrane bioreactors using anaerobic digester seed cultures, offering a sustainable and cost-effective strategy for nitrogen removal in engineered systems [133].

7. Factors Affecting Biological Gas Formation

The formation and emission of compounds such as H_2S , NH_3 , and VSCs are determined by a complex interaction of physical, chemical, and biological factors. These compounds not only have significant environmental impacts but also affect industrial processes, public health, and infrastructure quality. Understanding the factors influencing their formation is crucial for designing effective management strategies.

The formation of H_2S is significantly influenced by pH, temperature, and SRB activity. In anaerobic digestion of sewage sludge, higher initial pH values inhibit the activity of the relevant bacteria, significantly reducing H_2S production [134]. The release of H_2S during the drying of the thermal sludge is controlled by temperature, pH, and the active biomass present [135]. For example, higher temperatures, common in summer, accelerate microbial metabolic rates, leading to increased H_2S formation [136]. Similarly, prolonged retention times improve the interaction between microbes and substrates, facilitating the complete reduction of sulfates to sulfides [137]. In oceanic anoxic events, the sulfidation of organic matter can reduce H_2S concentrations in the water column by enhancing carbon preservation, highlighting the impact of geochemical processes in marine ecosystems [138]. Moreover, in industrial applications such as thiophene pyrolysis, the presence of water facilitates the formation of H_2S through hydrogen transfer and saturation of the thiophene ring [139].

The pH and temperature of the medium mainly influence the volatilization of NH_3 . In acidic or neutral environments, NH_4^+ is the predominant form, while in alkaline conditions, gaseous NH_3 volatilizes rapidly [107,140]. Ambient temperature, relative humidity, and manure management practices significantly influence NH_3 emissions in crop and livestock systems. Studies have shown that NH_3 emissions increase during the warmer months in dairy farms due to elevated temperatures [141]. Similarly, in croplands, soil properties, fertilizer placement, and weather conditions have been shown to affect NH_3 volatilization rates [142].

In wastewater treatment plants, processes such as NH_3 oxidation and accumulation of nitrite influence the emission of nitrous oxide (N_2O), a by-product closely linked to NH_3 transformations. The concentration of dissolved oxygen and temperature play an important role in denitrification kinetics, affecting the amount of N_2O emitted [143].

The formation of VSCs, such as CH_3SH and DMDS, is controlled by temperature, pH, and substrate composition. In composting, VSC emission peaks occur during the mesophilic and pre-thermophilic phases, with the temperature being a determining factor [144]. The formation of VSCs is also influenced by the temperature of the wastewater, SO_4^{2-} concentrations, and COD levels [145]. In treatment plants, VSC emissions tend to be higher in primary clarifiers and during warmer months due to increased temperature and organic loading [109]. In biosolids, the emission pattern of VSCs changes with time, with H_2S being the dominant compound throughout storage [146]. Other factors influencing VSC production include the concentration of free nitrogen NH_3 , which can inhibit key enzymes involved in VSC conversion [147].

Generally, the biological processes that lead to the formation of H_2S , NH_3 , and VSCs share several determining factors, including pH, temperature, substrate composition, and redox conditions. These variables not only modulate the microbial metabolic pathways responsible for producing these compounds but also influence their volatilization and dynamics in different environments. The regulation of chemical speciation and enzymatic activity depends on pH, while temperature controls both the microbial metabolic rates and the physical properties of gases. Furthermore, the availability of compounds such as sulfates, ammonium, and organic matter directly impacts transformation pathways. This integrated approach demonstrates the interconnectivity of environmental and microbial factors, emphasizing the need for a multidimensional analysis to understand the conditions that facilitate or hinder the formation of these compounds.

Incorporating the earlier discussion on how factors such as pH, temperature, substrate composition, and redox conditions influence biological gas formation, it is pertinent to broaden the analysis by considering studies that illustrate the complex interactions between environmental and operational factors. Several investigations have documented that those

climatic conditions, including seasonal variations, precipitation, and relative humidity, along with microbial activity, play a pivotal role in the emissions of volatile compounds such as NH_3 , H_2S , and other VOCs. The following text builds on this foundation, providing additional evidence that temperature and other environmental parameters act synergistically to modulate these processes, thereby deepening our understanding of gas formation dynamics in various real-world contexts.

Temperature emerges consistently as a critical factor across diverse environmental contexts. In wastewater treatment systems and composting processes, increased ambient temperatures during warm and dry conditions notably enhance the emissions of VSCs. For example, a study on aerobic composting reported significantly higher total VSC emissions of 561.89 mg/dry kg during summer compared to 358.45 mg/dry kg in spring, and only 215.52 mg/dry kg in winter [148]. Similarly, sequencing batch reactors exhibited elevated emission factors for H_2S ($361 \pm 101 \mu\text{g}/\text{ton}$) and methyl mercaptan ($82 \pm 76 \mu\text{g}/\text{ton}$) under higher temperatures [149].

Field studies across multiple climates provide further evidence supporting temperature's pivotal role. For instance, investigations at municipal waste biogas plants demonstrated significant correlations between concentrations of VOCs and NH_3 with temperature and relative humidity [150]. Likewise, at landfill sites, increased H_2S and NH_3 concentrations were closely associated with higher temperatures, air pressure variations, and wind direction, intensifying odor pollution under warm, humid conditions [151]. Additionally, odorant VOC emissions from apple pomace waste were found to increase by approximately 30% under open-air and oxygen-free storage conditions, further underscoring the sensitivity of biological gas formation to ambient environmental parameters [152].

Seasonal patterns, however, illustrate a moderating influence of cooler and wetter conditions. During rainy seasons, lower temperatures and increased precipitation typically result in reduced overall VSC production. Observations from wastewater treatment plants and natural sediment environments consistently report a reduction in VSC concentrations during wet periods, primarily due to decreased microbial activity rates and modified environmental chemistry [145,148]. Studies conducted in Bogotá further highlighted this seasonal contrast, recording H_2S concentrations peaking above 100 ppm during dry seasons, significantly surpassing levels during wetter periods characterized by lower temperatures and higher precipitation [153]. Table 3 presents the key characteristics and emission patterns of volatile sulfur compounds (VSCs) during dry seasons.

Table 3. Seasonal patterns in VSC formation: dry season characteristics. This table summarizes studies on seasonal patterns in volatile sulfur compound (VSC) formation, highlighting dominant compounds, concentration ranges, and key formation factors observed specifically during the dry season.

Study	Dominant VSCs	Concentration Ranges	Key Formation Factors
[154]	H_2S , MeSH	H_2S : 7 to 39,000 g/m ³	Organic matter, pH, Fe content
[148]	DMDS, DMS	Total VSCs: 561.89 mg/dry kg (summer)	Higher ambient temperatures
[149]	H_2S , MT	H_2S : $361 \pm 101 \text{ g}/\text{ton}$, MT: $82 \pm 76 \text{ g}/\text{ton}$	Higher ambient temperatures, aeration
[155]	MT	MT: 4 to 40 nM in sediment	Anaerobic mineralization, H_2S concentration
[156]	H_2S	Maximum 250 ppm	Higher ambient temperatures
[157]	DMS, CS_2 , DMDS	DMS: 608.5 g/m ³ , CS_2 : 658.5 g/m ³ , DMDS: 857.8 g/m ³	Higher ambient temperatures

The seasonal variability observed in volatile sulfur compound (VSC) concentrations (Table 3) suggests a strong influence of environmental conditions, particularly during the

dry season. To further explore this phenomenon, it is essential to examine the underlying mechanisms driving VSC formation. Table 4 presents the primary pathways responsible for generating these compounds, highlighting key contributing factors and their sensitivity to seasonal changes, providing deeper insight into the processes that result in the distinct dry season characteristics previously discussed.

Table 4. This table summarizes primary VSC formation pathways, highlighting contributing factors, seasonal influences, and possible control measures associated with each mechanism.

Formation Mechanism	Contributing Factors	Seasonal Influence	Control Measures
Anaerobic degradation	Organic matter content, sulfate concentration	Higher in warmer seasons	Aeration, chemical dosing
Methylation of H ₂ S	H ₂ S concentration, methyl donors	Temperature dependent	Sulfide control, pH adjustment
Biological sulfate reduction	Sulfate concentration, organic substrate	Enhanced in warmer temperatures	Sulfate removal, bioaugmentation
Thermal decomposition of sulfur compounds	Temperature, sulfur-containing organics	More significant in dry, hot seasons	Temperature control, feedstock management

Interaction among multiple factors also significantly influences gas formation. Research indicates that nitrogen fertilizer application rates, soil moisture, and temperature interact synergistically to promote NH₃ emissions, particularly in alkaline soils where the fertilizer rate exerts the strongest influence [158]. Elevated temperatures coupled with reduced soil moisture have been shown to amplify NH₃ volatilization, particularly pronounced in summer cropping systems [159,160].

Additionally, research in tropical wetland ecosystems emphasizes the complexity of factor interactions affecting VSC generation. In tropical coastal mangrove sediments and peatlands, gas emissions were significantly influenced by chemical parameters such as pH, salinity, redox potential, and organic substrate availability, as well as microbial community composition [88,161]. In peatland ecosystems, substrate additions rich in thiol and methylthiol groups led to enhanced H₂S, methanethiol, and dimethyl sulfide emissions. This effect was modulated by microbial communities, particularly sulfur-cycling bacteria like *Acidobacteria* and *Desulfosporosinus* [88]. Concurrently, in mangrove sediments, lower salinity, more electronegative redox potentials, and higher organic matter availability during the rainy season significantly favored methane production, demonstrating a clear seasonal dependency linked to environmental chemistry [161].

Moreover, adverse climatic conditions interacting with pollutants have been reported to enhance the bioavailability and bioaccumulation of contaminants, exacerbating oxidative stress and accelerating energy consumption in organisms [162]. Such environmental interactions underline the complexity inherent to biological gas production and emissions, highlighting the need for comprehensive analyses considering the multifactorial and synergistic effects of environmental variables.

8. Linking Water Quality and Malodorous Gas Emissions

8.1. Effects of Contaminants and Dissolved Oxygen on Odor Formation

The formation of malodorous gases in polluted water bodies is intrinsically linked to the availability of dissolved oxygen (DO) and the abundance of organic matter [163].

The presence of contaminants in water bodies usually results in low levels of DO due to the high oxygen demand exerted by organic pollutants and microbial activity [164]. The introduction of organic matter, such as agricultural runoff, untreated wastewater, and decaying vegetation, initiates the degradation process by aerobic microorganisms, which consume oxygen during this process [165]. This phenomenon, known as biochemical oxygen demand (BOD) [166], is one of the main factors in oxygen depletion. As the concentration of DO drops below critical thresholds, anaerobic processes dominate, creating conditions that promote the formation of gases such as H_2S , NH_3 , and VSCs [163].

The accumulation of organic matter in contaminated aquatic environments exacerbates oxygen depletion [165]. In environments with elevated concentrations of nutrients, such as nitrogen and phosphorus, eutrophication occurs as a result of human activities such as fertilizer use and discharge of untreated sewage [167]. A common consequence of eutrophication is the formation of algae blooms, which initially increase oxygen levels through photosynthesis [168]. However, as algae die and decay, microbial decomposition of organic material consumes large amounts of oxygen, often creating hypoxic or anoxic conditions. This leads to the formation of sulfides and dissolved iron, which result in malodorous black water agglomerates [39]. These oxygen-deficient zones promote the proliferation of SRB, such as *Desulfobacterota*, which metabolize SO_4^{2-} to H_2S [169]. Furthermore, the decomposition of aquatic plants, such as *Zizania latifolia*, contributes to the production of VSCs, including DMS, which exacerbates the odoriferous conditions [170].

The presence of contaminants in water bodies directly impacts the type and intensity of gas emissions. Pollutants such as nitrogen compounds, sulfates, and organic carbon serve as substrates for microbial activity [171,172], driving the production of malodorous gases. For example, elevated levels of dissolved organic carbon (DOC) in wastewater can enhance H_2S production during denitrification processes [173]. The decomposition of organic matter under anaerobic conditions in landfill leachates or swine slurry releases significant amounts of H_2S , a major contributor to malodors [174].

Inorganic pollutants, including heavy metals, also influence gas emissions by modifying the dynamics of the microbial community. For example, studies have shown that ferruginous sediments promote higher concentrations of DMS, while sulfidic sediments increase CH_3SH production [100]. These geochemical variations illustrate the complexity of malodorous gas formation in contaminated aquatic systems. Furthermore, the presence of fecal matter introduces both organic and microbial pollutants that exacerbate malodor formation. Sanitary inspections and tests H_2S have demonstrated a strong correlation between fecal contamination and H_2S emissions, highlighting the need for effective management of point sources of pollution [175].

Malodorous emissions are indicative of broader environmental degradation. Eutrophication of waterways caused by unregulated industrial and agricultural discharges contributes significantly to cyanobacterial blooms, which in turn release VSCs, leading to odor crises in drinking water systems [176]. Similarly, chlorination disinfection processes can result in the production of odor-causing substances, such as haloanisoles, in water supplies. This phenomenon complicates the management of water quality [177]. As water quality declines, the combined effects of microbial activity, organic matter accumulation, and pollutant interactions lead to the sustained release of malodorous gases, with notable ecological and public health consequences.

8.2. Real-World Data from Different Contaminated Water Sources: Case Studies from Tianziling Landfill and the Wuxi Crisis

The relationship between water quality and malodorous gas emissions can be further understood through real-world cases, which clearly illustrate the interaction between contamination and odorous emissions. While typically studied in riverine systems, other

aquatic environments or waste repositories can exhibit analogous conditions suitable for investigation.

For instance, although not a river system, the Tianziling landfill located north of Hangzhou city, China, represents a relevant case due to its similarity to river systems receiving untreated urban wastewaters. Tianziling was China's first standardized valley-type sanitary landfill, operational from 1991 to 2007 for its initial site, followed subsequently by a second landfill [151]. It had a designed filling capacity of approximately $2.2 \times 10^7 \text{ m}^3$, accepting exclusively municipal solid waste (MSW), excluding medical, industrial, or hazardous wastes. The landfill handled between 2000 and 4000 tons of MSW per day, accounting for over 60% of Hangzhou's municipal waste production [151]. The malodorous gases emitted from Tianziling landfill primarily resulted from the anaerobic decomposition of MSW, generating numerous volatile organic compounds (VOCs) and inorganic gases. Despite comprising less than 1% of total gaseous emissions, these odorants exhibited disproportionately adverse impacts due to their unique physicochemical characteristics. Notably, hydrogen sulfide (H_2S) concentrations ranged from 56.58 to 579.84 $\mu\text{g}/\text{m}^3$, and ammonia (NH_3) concentrations varied between 520 and 4460 $\mu\text{g}/\text{m}^3$. These two compounds significantly contributed to total odorant concentrations, with NH_3 accounting for approximately 83.91–93.94%, and H_2S representing 4.47–10.92% [151].

Similarly, the drinking water crisis in Wuxi, China, in 2007 exemplifies the direct impact of water quality deterioration on malodorous emissions in aquatic systems. During this crisis, nearly two million residents were affected by tap water exhibiting unpleasant odors and discoloration. The odor originated from Lake Taihu, caused predominantly by volatile organic sulfur compounds (VOSCs), including methyl thiols (MeSH), dimethyl sulfide (DMS), dimethyl disulfide (DMDS), and dimethyl trisulfide (DMTS) [178]. These VOSCs were generated through microbial decomposition following a massive cyanobacterial bloom triggered by illegal industrial discharge and inadequately regulated domestic pollution, driving the lake ecosystem towards a critical state [178]. Chemical analysis revealed thiols and thioethers as the primary odor contributors, with an initial dominance of MeSH and DMS, subsequently oxidized to DMDS and DMTS as dissolved oxygen levels increased. Although other potentially odorous compounds such as β -cyclocitral, geosmin, and 2-methylisoborneol (2-MIB) were detected, their limited concentrations or higher odor thresholds indicated minor roles in odor generation during this specific crisis [178]. An odoriferous "black water agglomeration" formed at the water intake point, presumably from large-scale releases of VOSCs resulting from decomposed cyanobacteria and lake sediments. Emergency treatment measures including potassium permanganate oxidation and powdered activated carbon adsorption were effectively implemented to mitigate odor issues in tap water [178].

Both cases emphasize the critical linkage between deteriorating water quality conditions and significant malodorous emissions, underlining the necessity of integrated management strategies to control and mitigate such environmental challenges.

8.3. Case Study: The Bogotá River

The Bogotá River starkly illustrates the environmental consequences of unregulated urbanization, industrial growth, and agricultural intensification. Spanning 380 km from its source in the Páramo de Guacheneque to its confluence with the Magdalena River, this vital waterway is among the most contaminated in Colombia [179]. The upper and middle basins of the Bogotá River, which traverse highly urbanized and industrialized areas, are subject to severe pollution caused by the discharge of untreated domestic and industrial wastewater, the discharge of agrochemicals, and the disposal of solid waste [180,181].

These activities have led to a significant reduction in DO levels and an increase in organic matter content, creating conditions conducive to the formation of malodorous gases and other indicators of poor water quality. The main sources of organic and chemical pollutants in the Bogotá River are domestic and industrial discharges. The wastewater treatment infrastructure in Bogotá and the surrounding municipalities is inadequate to handle the volume of effluents generated, resulting in the direct release of untreated waste into the river [179]. This has led to levels of microbial contamination that are considered unacceptable. In particular, the concentration of total coliforms and *Escherichia coli* has been observed to exceed the permissible limits, especially in the middle basin [181]. Industrial discharges introduce additional contaminants, including heavy metals, perfluoroalkyl and polyfluoroalkyl substances (PFAS), and complex organic compounds. PFAS concentrations are highest in areas with intense industrial activity, such as near major factories and population centers [182]. These substances persist in the environment and pose long-term risks to human and ecological health [183]. Furthermore, river sediment samples reveal a significant accumulation of pollutants, acting as reservoirs that perpetuate contamination and release odorous compounds into the water column over time [180].

The organic matter content of the river is further exacerbated by agricultural runoff, particularly in the upper basin, where the excessive use of agrochemicals contributes to the load of nutrients [184]. This enrichment of nutrients triggers eutrophication, leading to algal blooms that consume oxygen during decomposition, compounding the problem of oxygen depletion of the river [185]. These processes create a feedback loop, as the decaying biomass becomes an additional source of organic material and gases such as NH_3 and H_2S further degrade the quality of the water. One of the most critical issues facing the Bogotá River is the consistently low dissolved oxygen levels. Studies from 2008 to 2017 have classified water quality as regular to poor in the upper basin, with only five monitoring stations reporting acceptable conditions during this period [180]. The primary driver of oxygen depletion is the high BOD exerted by organic pollutants. Large volumes of untreated domestic and industrial effluents contribute substantial organic matter, which microorganisms metabolize, consuming oxygen in the process [186]. This imbalance leads to hypoxic or anoxic conditions, which are incompatible with healthy aquatic ecosystems [181].

In the middle basin, where the river receives most of its pollutant load, dissolved oxygen levels are critically low. Discharges can reduce oxygen concentrations by $20 \mu\text{mol/L}$ in rivers [187]. High levels of organic carbon, nutrients, and other contaminants from domestic wastewater further exacerbate oxygen depletion [188]. These conditions not only limit the river's ability to support aquatic life, but also promote anaerobic microbial processes. SRB thrive in such conditions, metabolizing sulfates and organic matter to produce H_2S and VSC, responsible for the notorious malodors of the river [189].

The issue of unpleasant odors resulting from contamination in the Bogotá River clearly highlights the urgent need for comprehensive solutions aimed at improving water quality in this aquatic system. Effective water quality improvement must begin at the sources of contamination, making it essential to control polluted water discharges originating from both rural and urban areas through which the river flows. Several strategies have been proposed for this purpose, including optimizing and redesigning the water quality monitoring network to enhance sampling efficiency and reduce operational costs [190]. Additionally, the implementation of cleaner production practices along with the economy principles has been emphasized to minimize waste generation and reduce environmental impacts from industrial and urban activities [191].

Another significant solution involves utilizing water hyacinth (*Eichhornia crassipes*) as a natural method for organic matter degradation, nutrient absorption, and heavy metal removal in affected water bodies, such as the Muña Reservoir [192]. Furthermore, urban

water management strategies, including stormwater harvesting, industrial wastewater reuse, and adoption of water-saving technologies in residential areas, have been evaluated. Simulations of these strategies indicate potential significant reductions in drinking water consumption and wastewater generation, particularly in the urban catchments of the Fucha and Tunjuelo rivers [193].

These integrated initiatives aim to mitigate the negative impacts caused by rapid urban growth, reduce the formation of odor-causing compounds, and substantially contribute to the restoration and preservation of the environmental quality of the Bogotá River and its tributaries.

9. Detection Technologies for Odorous Gases

9.1. Traditional Analytical Techniques

Various measurement protocols have been developed for the effective analysis of odor compounds. Among these, gas chromatography coupled with sulfur chemiluminescence detection (GC-SCD) is widely used. This approach, which incorporates N_2O as an internal standard, has shown low detection limits and high reproducibility, making it particularly effective for the detection of compounds such as CH_3SH , DMS, and DMDS [194]. The combination of thermal desorption with GC-SCD allows the analysis of a wider range of VSCs present in sewer headspace air, including H_2S and ethanethiol [64]. In ambient measurements, cryotrapping-enhanced GC-SCD proves effective in detecting TRS at concentrations as low as 120 ppt in 100 mL of air, though its sensitivity diminishes for H_2S and CH_3SH below 2 ppb [195]. Similarly, olfactory detection mass spectrometry has been instrumental in identifying VSCs of biosolid emissions, shedding light on shifting emission patterns as biosolids age [146]. In addition to these techniques, thermal desorption followed by gas chromatography–mass spectrometry (GC-MS) has been used to assess VSCs in air samples collected from sewage management areas. In these samples, DMS, carbon disulfide and DMDS were identified as the predominant compounds [157].

For air odor measurement, these methods integrate sensory and chemical approaches, each contributing strengths to compound characterization. Among chemical methods, thermal desorption combined with gas chromatography and sulfur chemiluminescence detection (TD-GC-SCD) can measure multiple VSCs, such as H_2S and CH_3SH , in air in the sewer headspace with detection limits ranging from 0.10 to 5.26 $\mu g/m^3$ [64]. The combination of TD-GC-MS with simultaneous olfactory evaluation has proven effective in the identification and quantification of odorous compounds from various sources. This method detects trace amounts of compounds at concentrations up to 40 times higher than those typically found in air [196]. A novel quantification approach using TD-GC-MS has shown remarkable detection limits for numerous key odorants [197]. Air dispersion models provide valuable tools to assess spatial and temporal odorant distribution and to assess potential mitigation strategies [198].

Furthermore, UPLC-MS methods for the analysis of NH_3 have achieved submicromolar detection limits [199], and satellite measurements of NH_3 have been used to estimate H_2S exposure from animal husbandry operations by integrating remote sensing with in situ data [200]. Spectroscopic techniques, such as hyperspectral infrared imaging, have revealed underestimated industrial sources and demonstrated the potential of future satellite sounders to improve monitoring accuracy [201]. UV-visible chemiluminescence spectroscopy has been effective in examining ammonia–hydrogen–air flames, facilitating nonintrusive assessments of equivalence ratios and fuel fractions [202]. In agricultural applications, open-path tunable diode laser absorption spectroscopy has enabled high temporal resolution NH_3 measurements, elucidating meteorological influences and nitrogen application rates [203].

To clearly illustrate the differences among these analytical methodologies regarding cost, sensitivity, and practicality, a comparative summary table is presented (Table 5).

Table 5. Comparative summary of traditional analytical detection technologies for odorous gases based on cost, sensitivity, and practicality.

Method	Cost	Sensitivity	Practicality	References
GC-SCD	High	120 ppt–2 ppb	Laboratory-based, specialized equipment	[64,194,195]
TD-GC-SCD	High	0.10–5.26 $\mu\text{g}/\text{m}^3$	Laboratory-based, broader VSC analysis	[64]
TD-GC-MS	High	Trace level ($\mu\text{g}/\text{m}^3$ to sub-ppb)	Laboratory-based, rapid, precise, integrated olfactory evaluation	[157,196,197,204,205]
UPLC-MS	High	Submicromolar (below μM range)	Laboratory based, rapid, accurate, ideal for complex samples	[199,206,207]
Satellite Remote Sensing	Moderate–High	Indirect $\text{NH}_3/\text{H}_2\text{S}$ estimation	Large-scale coverage, good for regional assessments but indirect measurement	[200,201]
UV-visible Chemiluminescence	Moderate–High	High (ppb–ppm range)	Specialized instrumentation, limited field portability	[202]
Open-path Tunable Diode Laser Absorption	Moderate–High	High temporal resolution (ppb)	Excellent field applicability, good temporal resolution, field deployable	[203]
Air Dispersion Models	Moderate	Variable (depends on input data)	Computational approach, requires input data, useful for scenario modeling	[198]

9.2. Advanced Optical and Spectroscopic Methods

For simultaneous detection of H_2S and SO_2 , a portable mid-infrared sensor that employs substrate-integrated hollow waveguides and UV conversion has demonstrated detection limits as low as 207 ppbv for H_2S [208]. Cavity-enhanced near-infrared laser absorption spectroscopy is another option that offers rapid and precise measurements of H_2S in fuel reforming streams with minimal cross-interference [209]. Further advances in mid-infrared sensors using substrate-integrated hollow waveguides have improved the applicability of ppb-level detection of both H_2S and SO_2 [208].

Quartz-enhanced photoacoustic spectroscopy (QEPAS) in conjunction with external cavity quantum cascade lasers and hollow-core waveguides has demonstrated detection limits of 450 ppb for H_2S in just three seconds [210]. Recent innovations in QEPAS, such as dual resonance photoacoustic spectroscopy and blocking of laser cavity molecules, have reduced the detection limits of H_2S to 10 ppb [211].

Cavity ring-down spectroscopy (CRDS) has demonstrated excellent sensitivity with minimal interference from humidity or volatile organic compounds [212]. CRDS and other laser-based techniques are ideal for real-time, continuous monitoring of odorants in various environmental settings, offering an avenue to achieve more effective and rapid odor control.

Table 6 presents a comparative summary of these advanced optical and spectroscopic detection technologies, clearly illustrating their relative cost, sensitivity, and practical considerations.

Table 6. Comparative summary of advanced optical and spectroscopic detection technologies for odorous gases based on cost, sensitivity, and practicality.

Method	Cost	Sensitivity	Practicality	References
Mid-infrared Sensor (Hollow Waveguide with UV)	High	~207 ppbv (H ₂ S, SO ₂)	Portable, rapid, suitable for simultaneous multi-gas detection	[208,208]
Cavity-enhanced Near-infrared Laser Absorption Spectroscopy	High	ppbv level	Fast response, minimal interference, precise for fuel streams	[209]
Quartz-enhanced Photoacoustic Spectroscopy (QEPAS)	High	10–450 ppb	Ultra fast, highly sensitive, suitable for real-time analysis, minimal cross-interference	[210,211]
Cavity Ring-Down Spectroscopy (CRDS)	High	ppb level	Real-time monitoring, highly sensitive, minimal VOC and humidity interference	[212]

9.3. Electrochemical Sensors and Colorimetric Methods

Advances in electrochemical sensors for sulfur compounds have yielded promising results, offering improved sensitivity and faster detection capabilities. Electrochromic compounds, such as 4,4',4'',4'''-tetrakis(dimethylamino)-tetraphenylethylene, can detect both NO₂ and SO₂ in the gas phase, with the detection of SO₂ enhanced by the presence of t-butylhydroperoxide [213]. For H₂S, triple-pulse amperometry enables direct electrochemical sensing without sulfur poisoning, achieving high sensitivity and rapid response times [214]. Electrochemical Impedance Spectroscopy (EIS) has emerged as a highly sensitive technique for detecting toxic gases, including H₂S, using semiconducting metal oxides [215]. Thin-layer amperometric sensors with microelectrodes have been developed, providing membrane-independent responses and improving the practicality of Clark-type sensors for monitoring H₂S [216].

Similarly, flame photometric detectors (FPDs) have significantly improved the detection of VSCs by reducing detection limits and enhancing performance. A multiple flame FPD (mFPD) demonstrated resilience to hydrocarbon quenching and achieved detection limits as low as 4×10^{-11} g S/s [217]. A quartz rod-enhanced FPD (qFPD) set a new benchmark with detection limits of 0.3–0.5 pg S/s, outperforming conventional detectors by a factor of 5–7 [218]. Microcountercurrent FPD (microcc-FPD) enabled ultrafast gas chromatographic separations with superior performance in detecting narrow peaks, achieving detection limits of 3 ng S/s for organosulfur compounds [219]. Pulsed FPD (PFPD) further advanced these developments with detection limits between 2.37 and 4.89 pg for various reduced sulfur compounds, maintaining consistent performance at concentrations above $\sim 20 \text{ nmol mol}^{-1}$ [220].

For NH₃ analysis, ref. [199] introduced a UPLC-MS method utilizing dansyl chloride derivatization, achieving submicromolar detection limits. Ref. [200] employed satellite

NH₃ measurements to estimate H₂S exposure from animal husbandry operations, integrating remote sensing with in situ data. Spectroscopic techniques have also been crucial in addressing NH₃ emissions, such as hyperspectral infrared imaging to detect industrial sources [201] and ultraviolet-visible chemiluminescence spectroscopy to examine ammonia–hydrogen–air flames [202]. In agricultural applications, open-path tunable diode laser absorption spectroscopy has enabled high temporal resolution measurements of NH₃ [203].

Colorimetric and smartphone-assisted detection techniques have emerged as promising alternatives to low-cost, portable monitoring of NH₃ and H₂S. Ref. [221] developed a 3D-printed platform with paper sensors for simultaneous detection of NH₃ and sulfide in wastewater. Ref. [222] introduced QRsens, a dual-purpose QR code with built-in colorimetric sensors for in situ air analysis. Nanocatalysts have enabled ultrasensitive colorimetric detection of dissolved H₂S, with detection limits of 7.5 nM [223]. Smartphone-based methods have also been refined; for example, ref. [224] used silver nanoparticles to detect NH₃ with a detection limit of 200 mg/L.

Table 7 summarizes these electrochemical, colorimetric, and other emerging detection technologies clearly, comparing their cost, sensitivity, and practicality explicitly.

Table 7. Comparative summary of electrochemical sensors, colorimetric methods, and emerging detection technologies for odorous gases based on cost, sensitivity, and practicality.

Method Category	Method	Cost	Sensitivity (Detection Limit)	Practicality	References
Electrochemical Sensors	Triple-pulse amperometry	Moderate–Low	ppb to ppm levels	Portable, rapid detection, resistant to sulfur poisoning	[214]
	Electrochemical impedance spectroscopy	Moderate–Low	ppb levels	Highly sensitive, suitable for toxic gases, portable	[215]
Flame Photometric Detectors (FPD)	Multiple flame FPD (mFPD)	High	4×10^{-11} g S/s	High sensitivity, resistant to interference	[217]
	Quartz rod-enhanced FPD (qFPD)	High	0.3–0.5 pg S/s	Exceptionally sensitive, stable signal	[218]
	Microcountercurrent FPD (microcc-FPD)	High	3 ng S/s	Ultrafast chromatographic separations, high precision	[219]
	Pulsed FPD (PFPD)	High	2.37–4.89 pg	Excellent performance, stable at varying concentrations	[220]
Colorimetric and Smartphone Methods	3D-printed paper sensors	Low	ppm levels (NH ₃ , H ₂ S)	Affordable, portable, suitable for rapid field analysis	[221]
	QR code-integrated sensors (QRsens)	Low	ppm levels (NH ₃ , H ₂ S)	Quick field measurements, accessible via smartphones	[222]
	Nanocatalyst-enhanced assays	Moderate–Low	7.5 nM (H ₂ S)	Ultrasensitive, rapid response, field compatible	[223]
	Smartphone-based silver nanoparticles	Low	200 mg/L (NH ₃)	User friendly, portable	[224]

9.4. Emerging Technologies

Recent advances in emerging technologies have significantly enhanced the detection capabilities for hazardous gases such as NH₃, H₂S, and volatile sulfur compounds (VSCs), highlighting the potential of artificial intelligence (AI)-based methods to improve accuracy, sensitivity, and efficiency in environmental monitoring.

In agricultural contexts, cavity ring-down spectroscopy (CRDS) has demonstrated excellent sensitivity for precise NH₃ detection [193]. Additionally, wind tunnels calibrated

to ambient wind speeds provide reliable ammonia emission data [194]. Cost-effective methods such as passive flux samplers combined with dispersion models show promise in multipoint experimental setups [195]. However, an intercomparison involving 13 instruments highlighted the urgent need for standardized protocols regarding inlet design, calibration, and maintenance to ensure reliable data in routine monitoring [146].

The integration of smart sensors and Internet of Things (IoT) technologies has substantially advanced the detection of harmful gases such as H_2S . Chemiresistive sensors based on metal oxide semiconductors, particularly those utilizing nanostructures like nanofibers, exhibit high surface area and excellent electrical conductivity, facilitating effective H_2S detection [225]. Novel fabrication methods, such as Pd-functionalized CuCrO_2 thin films, have achieved detection capabilities for H_2S concentrations as low as 0.5 ppm [226]. The integration of these sensors with IoT technologies enables real-time monitoring and wireless data transmission, significantly improving safety in industrial and environmental applications [227]. Overall, these emerging technologies carefully consider critical factors such as sensitivity, selectivity, and response time to ensure effective H_2S monitoring in petroleum refineries, sewage treatment plants, and fuel cell systems [228].

Additionally, electronic noses (E-noses)—artificial olfaction systems consisting of gas sensor arrays, sample handling, and signal processing components—have gained considerable attention [229]. When combined with artificial intelligence algorithms, E-noses efficiently detect and discriminate various gases, significantly contributing to improved environmental monitoring, industrial safety, and medical diagnostics [230]. Recently, ultra-low-power E-noses based on three-dimensional tin oxide nanotube arrays have been developed, achieving high sensitivity at room temperature and substantially reducing power consumption compared to conventional devices [231]. In agriculture, electronic noses demonstrate potential for detecting low ammonia concentrations typically emitted during farming activities, providing real-time monitoring and enhanced sensitivity compared to traditional methods [232].

Finally, colorimetric techniques combined with AI-based analysis have emerged as promising methods for detecting gases such as H_2S , NH_3 , and VSCs. Ref. [233] developed a dual-sensing system assisted by deep learning for rapid H_2S detection, covering a wide concentration range from 0.1 to 100 ppm. Ref. [221] implemented a smartphone-assisted method using paper sensors and a 3D-printed platform, enabling simultaneous monitoring of ammonia and sulfide in wastewater. Additionally, ref. [234] proposed a nylon-supported plasmonic assay utilizing silver nanoparticles, capable of detecting VSCs in breath samples with a detection limit of 45 ppbv for H_2S . Furthermore, ref. [235] developed paper-based colorimetric sensors for H_2S measurement in water and air samples, achieving detection limits as low as 1.11–1.12 mL/m^3 in air and 0.5 mg/L in water. These studies clearly highlight the potential of AI-assisted colorimetric techniques to substantially improve the rapid, sensitive, and efficient detection of critical environmental gases such as H_2S , NH_3 , and VSCs.

10. Advantages and Limitations of Different Measurement Methods

This section presents a comprehensive comparative analysis of the methodologies discussed earlier, with a particular focus on their respective strengths and limitations. Table 8 provides an overview of the techniques discussed, categorizing them according to their specific applications, operational advantages, and constraints. This facilitates a clear understanding of their suitability for various contexts.

Table 8. Comparative analysis of measurement techniques: applications, advantages, limitations, and color-coded effectiveness. A color-coded effectiveness column has been added. Green indicates high accuracy and sensitivity, making these methods especially suited for detecting low-concentration compounds. Orange represents moderate effectiveness, where methods may not reach the same low detection limits or selectivity but often offer practical benefits.

Methodology	Working Principle	Application	Advantages	Limitations	Effectiveness
GC-SCD	Gas chromatography coupled with sulfur chemiluminescence detection.	H ₂ S, VSC	High selectivity and sensitivity in various matrices [236]. Suitable for low concentrations [195].	Requires advanced equipment prone to frequent maintenance and component corrosion [236]. Non-linear responses at low concentrations for some compounds [195].	High selectivity and sensitivity for sulfur-containing compounds; very low detection limits.
TD-GC-SCD	Gas chromatography with thermal desorption and chemiluminescence detection.	H ₂ S, VSC	Detects a wide range of sulfur compounds [237]. High sensitivity [64].	For complex matrices like straight-run gas oils, high-resolution mass spectrometry may be necessary [238].	Similar to GC-SCD, optimized for volatile compounds using thermal desorption; excellent for trace-level sulfur compounds.
GC-MS	Gas chromatography coupled with mass spectrometry.	H ₂ S, VSC	Qualitative and quantitative analysis capabilities [239]. Suitable for odor studies with simultaneous olfactory evaluation [196].	Long analysis times [240]. Requires specialized equipment and expertise [241].	Broad-spectrum identification and quantification; high precision.
TD-GC-MS	Gas chromatography with thermal desorption and mass spectrometry.	H ₂ S, VSC	Optimized detection limits for trace volatile organic compounds [204]. Short collection times [205].	Susceptible to contamination risk [242]. May be more affected by matrix effects than other methods [204].	Enhanced sensitivity for highly volatile compounds; short sampling times.
Laser Spectroscopy (NIR, QEPAS)	Measurement based on laser light interaction with gaseous compounds, enhancing sensitivity with optical cavities.	H ₂ S, NH ₃	Real-time continuous monitoring [243]. High sensitivity and low detection limits [210].	Gas flow and temperature effects at high excitation powers [244]. Requires frequent calibration and long response times [243].	Capable of ppb or sub-ppb detection in real time; ideal for continuous monitoring of one/few specific compounds.
Electrochemical Sensors (EIS, Amperometric)	Based on detecting changes in current or impedance in response to gas presence.	H ₂ S, NH ₃	Compact, portable, and cost-effective [245]. Ease of operation and simplicity of construction [245].	Sensitive to electrode contamination [246]. Potential for nonspecific impedance changes [246].	Portable and inexpensive, but less accurate and prone to drift under complex conditions.
FPD and mFPD Detectors	Based on light emission from flames interacting with sulfur compounds.	H ₂ S, VSC	High sensitivity for sulfur compounds [219]. Low detection limits [219].	Requires controlled combustion conditions, which can be challenging in unstable systems [247]. Flame stability issues in certain environments [247].	High sensitivity to sulfur species; excellent detection limits, though they require stable flame conditions.
UPLC-MS	Ultra-performance liquid chromatography coupled with mass spectrometry for NH ₃ .	NH ₃	High precision for volatile nitrogenous compounds [206]. Rapid analysis with high sensitivity [207].	Detection of other compounds may obscure the detection of NH ₃ [207].	Highly accurate for volatile nitrogen compounds (especially ammonia); very fast and sensitive. Less common for volatile sulfur.
TDLAS	Tunable diode laser absorption spectroscopy to measure spectral absorption of specific gases.	NH ₃ , H ₂ S	High precision in monitoring continuous emissions [248]. Suitable for harsh environmental conditions [248].	Inherent sensitivity limitations due to infrared absorption principles [249].	Excellent for continuous measurements, high sensitivity, robust in industrial environments.

Table 8. Cont.

Methodology	Working Principle	Application	Advantages	Limitations	Effectiveness
Colorimetric Techniques and Smartphone Detection	Color changes induced by chemical reactions detected via optical platforms like smartphone cameras.	NH ₃ , H ₂ S	Affordable and portable technology [221]. Suitable for rapid, low-cost measurements [221].	Limited selectivity and detection limits compared to laboratory instruments [250]. Detection limits ranging from 1 to 200 mg/L for NH ₃ and H ₂ S [224].	Very accessible and portable; lower selectivity and detection limits compared to advanced instrumentation.
CRDS (Cavity Ring-Down Spectroscopy)	Laser absorption through multiple reflections in a closed cavity for NH ₃ and H ₂ S analysis.	NH ₃ , H ₂ S	Ideal for real-time monitoring with high sensitivity [251]. Excellent linearity over a wide range of NH ₃ concentrations [212].	Slight humidity effects at extreme conditions [212].	Highly sensitive real-time measurements; ideal for trace gases.
Wind Tunnels and Passive Samplers	Devices that capture gases in open fields analyzed using dispersion models.	NH ₃	Cost-effective and easy to implement over large areas [252]. High precision and fast response [253].	Dependent on climatic conditions such as wind speed and temperature [252]. May underestimate NH ₃ concentrations under low wind speed conditions [254].	Useful to estimate emissions over large areas; heavily dependent on weather (wind speed, temperature). Less precise for low wind speeds.

The comparative analysis presented in Table 8 demonstrates the wide range of methodologies available to measure H₂S, VSCs, and NH₃. Although some methods demonstrate superior sensitivity and precision, others are distinguished by portability, cost effectiveness, or suitability for specific field conditions. The selection of the appropriate methodology is contingent on several factors, including the target compound, the environmental context, and analytical requirements. This comparison highlights the need for ongoing advances in measurement techniques to address emerging challenges in accuracy, cost effectiveness, and environmental adaptability.

Finally, it is essential to highlight that cost, required expertise, and environmental variability often pose significant barriers to implementing these measurement techniques in real-world scenarios. For instance, high-precision methods like GC-MS or CRDS, while offering excellent sensitivity, require expensive instrumentation and specialized training to operate and interpret data. Continuous monitoring systems such as TDLAS or QEPAS also demand frequent calibration and robust infrastructure, which can be challenging in remote areas or when budgets are tight.

In contrast, lower-cost approaches (e.g., electrochemical sensors, colorimetric kits) are more accessible and portable but may suffer from weaker selectivity or signal drift, especially under fluctuating temperature and humidity. Moreover, in industrial or field conditions, the presence of multiple interfering pollutants can compromise accuracy for even the most advanced systems. Ultimately, selecting the right method depends on balancing performance goals (like ultra-low detection limits) with practical feasibility (budget constraints, ease of use, and availability of trained personnel).

11. Mitigation and Control Strategies

11.1. Removal Challenges

The removal of gaseous compounds such as H₂S and NH₃ presents significant challenges due to their chemical properties and environmental and health implications. Although both are common in industrial gas streams and can cause serious pollution problems, difficulties in their removal vary based on their toxicity and the availability of treatment technologies.

H₂S is highly toxic, making its control and removal a top priority [255]. Its presence in industries such as oil and gas requires highly specialized methods to reduce concentrations to safe levels [256]. These methods, ranging from physical and chemical processes to biological treatments, have been significantly developed to address the hazardous nature of H₂S [19]. The industry has made considerable progress in its removal, prioritizing efficient solutions such as converting gas into less harmful compounds or capturing it using liquid and solid absorbents, while optimizing separation technologies [19].

In contrast, although less toxic, NH₃ presents significant challenges on its own [19]. The removal of NH₃ can often be more complex and costly than H₂S due to its high solubility in water and the potential for the formation of unwanted by-products during treatment [257]. In addition, the removal processes of NH₃ often require more careful control to avoid issues such as ammonium salt formation or other contaminants [258]. Although its presence is typically managed through biological or adsorption processes, the need for more specialized treatment technologies can make the removal of NH₃ more expensive and complicated, particularly in agricultural and wastewater contexts [259].

Therefore, while both gases present challenges to removal, NH₃ often requires more complex and costly treatment methods compared to H₂S, which benefits from the more widely established and developed technologies described in the next section.

11.2. Methods for H₂S Removal

The removal methods for H₂S can be classified according to the phenomenon they rely on: absorption, adsorption, membrane separation, and chemical conversion.

11.2.1. Absorption

Absorption is one of the most established and widely used mechanisms for the removal of H₂S in the oil and natural gas industry, using a liquid solvent as the capture medium [256]. Absorption is divided into two main categories: physical and chemical [260]. Physical absorption is based on the solubility of H₂S in the solvent, utilizing weak intermolecular interactions, which facilitates the subsequent release of the absorbed gas when conditions change [261]. A typical example of a physical solvent is sulfolane, which is used for its high capacity to dissolve acid gases such as H₂S without chemically reacting with them [262]. In contrast, chemical absorption involves the formation of strong chemical bonds between H₂S and the solvent, resulting in more efficient removal, although it requires more energy to regenerate the solvent [263]. The chemical solvents most commonly used for H₂S absorption are alkanolamines, such as monoethanolamine (MEA), diethanolamine (DEA), and methyl diethanolamine (MDEA), due to their effectiveness and widespread use in the oil and natural gas industry [264].

11.2.2. Adsorption

Adsorption is the process by which fluid molecules adhere to a solid surface [19]. Similarly to absorption, this mechanism is divided into two types: physical adsorption and chemical adsorption [260]. Physical adsorption is based on weak intermolecular interactions, such as van der Waals forces, between the fluid molecules and the adsorbent surface [265]. A common example is activated carbon, valued for its wide availability, low production cost, and ecological origin [266]. On the other hand, chemical adsorption involves chemical bonds [260]. In some cases, the adsorbent, such as activated carbon, can be impregnated with basic compounds such as sodium hydroxide or potassium hydroxide to promote chemical reactions with H₂S, converting the gas into stable compounds such as sulfides, thus improving the capture capacity of the adsorbent [267].

11.2.3. Membrane Separation

Membrane separation is well-established in various industrial processes [19]. Due to its reduced carbon footprint, high modularity, low weight, and ease of operation compared to absorption and adsorption, this technology is widely used in remote areas and offshore operations [268]. Membrane separation involves the use of a porous membrane through which selective gas transport occurs based on differences in the solubility and diffusion of different components through the membrane [260]. Polymeric membranes, such as polysulfone and polyamide, are the most commonly used materials due to their ease of fabrication, flexibility, and high effectiveness in separating gases such as carbon dioxide, oxygen, and H₂S [269].

11.2.4. Chemical Conversion

Chemical conversion is the process of transforming H₂S into other less dangerous compounds or, in some cases, useful compounds [260]. One of the most widely used conversion routes is the Claus Process, implemented in the natural gas industry and oil refineries [270]. The Claus Process converts H₂S into elemental sulfur through the partial combustion of H₂S in the presence of oxygen, followed by the catalytic reaction of the resulting gas mixture [271]. This sequence of reactions allows sulfur recovery with an efficiency exceeding 95% [272].

Another conversion route is liquid-phase catalytic oxidation, where H₂S is oxidized in the liquid phase, typically in the presence of a catalyst and an oxidizing agent, transforming it into sulfuric acid [260]. This method offers advantages such as high selectivity and conversion efficiency, being especially effective in applications where removal of H₂S is required in liquid streams, such as in industrial wastewater treatment or gas streams with high water vapor content [273]. The ability to handle large volumes of contaminated liquid makes this process particularly useful in sectors where H₂S streams are dissolved or mixed with other liquid components [274].

For processes that require an easily applicable method under standard conditions for the treatment of residual gases, packed-bed bioreactors, known as biofilters and biopercolators, are used for H₂S oxidation [275]. In these systems, a consortium of microorganisms oxidizes H₂S, converting it to elemental sulfur under anaerobic conditions or sulfuric acid under aerobic conditions [276]. Biofilters and biopercolators are highly effective and provide high conversion of H₂S due to the efficiency of microorganisms in the oxidation process [277].

Biofilter

Biofilters are widely used in industry for the treatment of residual gases due to their ability to efficiently remove volatile organic compounds and other gaseous pollutants [278]. These systems operate through a bed of organic material, such as compost or peat, which provides an environment suitable for the growth of microorganisms capable of using pollutants as an energy source, transforming them into other compounds that are easier to manage [276]. In industrial applications, biofilters are used to treat emissions from processes such as chemical manufacturing, food processing, and the paper industry [279]. Their use is valued for their ability to reduce odor levels and toxic gases, as well as for their lower operational cost compared to other gas treatment technologies [275].

Biopercolators

Biopercolators, or percolating biofilters, are a variant of the biofilter that operates as a three-phase bioreactor: solid, liquid, and gas [280]. This system not only filters and oxidizes gases such as H₂S but also allows the circulation and recirculation of water through the

packed bed, improving the hydration of the biofilm and facilitating the removal of oxidation products such as elemental sulfur or sulfuric acid [281]. Biopercolators are designed to operate under controlled conditions that optimize the efficiency of the biological process and are used in industrial applications where H₂S generation is constant and requires a more robust and efficient solution, such as wastewater treatment [275].

Comparison in Diverse Settings

In urban settings, high population density and low odor tolerance often favor end-of-pipe solutions (e.g., covers, scrubbers, or biofilters) that capture and treat gases before release [282]. Rural areas, especially where agricultural practices dominate, may rely on preventive measures—such as improved waste handling and controlled manure application—since full-scale industrial systems are not always economically feasible [283]. Industrial sites typically adopt advanced, technology-driven mitigation (e.g., catalytic oxidation, integrated chemical/biological treatments) to meet strict emissions targets, while natural or semi-natural environments lean toward ecological restoration (e.g., re-oxygenation, nutrient input reduction) [284]. In all cases, policy frameworks usually regulate the maximum allowable emissions rather than mandating a specific treatment method; thus, each facility can choose the most cost-effective or context-appropriate technology to comply with emission limits.

Table 9 summarizes the most commonly used processes for the removal of H₂S, and Figure 3 summarizes the diagrams of these processes.

Table 9. Comparative table between diverse H₂S elimination techniques.

H ₂ S Removal Method	Principle	Medium Used	Advantages	Disadvantages
Absorption	Removal through solubility or reaction capability in a liquid medium.	Physical: sulfolane. Chemical: alkylamines.	High efficiency in systems with high gas flow rates.	Requires solvent regeneration.
Adsorption	Separation by attachment to the surface of a solid through weak intermolecular interactions.	Physical: activated carbon. Chemical: activated carbon impregnated with strong bases (NaOH, KOH).	Low operating cost. Wide availability of adsorbents.	Adsorbent saturation requires frequent replacement. Chemical adsorption implies additional costs.
Membrane Separation	Segregation by forcing passage through a porous membrane due to differences in solubility and diffusion.	Polymeric membranes: polysulfones and polyamides,	High modularity and ability to operate in remote areas. Low environmental impact.	Lower efficiency compared to other methods. High installation costs.
Claus Process	Conversion to elemental sulfur through partial combustion and the catalytic reaction of resulting products.	Oxygen, catalyst (Al ₂ O ₃).	High recovery of elemental sulfur.	Requires controlled operating conditions. Secondary emissions need to be controlled.
Liquid Catalytic Oxidation	Conversion to sulfuric acid in a liquid medium using an oxidizing agent and a catalyst.	Oxidizing agent (O ₂), catalyst (Fe ₂ O ₃).	High selectivity and efficiency in liquid streams.	Risk of corrosion. Difficult operation handling.
Biofilter	Conversion to sulfuric acid or elemental sulfur through aerobic biological oxidation by sulfur-oxidizing microorganisms.	Packed-bed bioreactor.	Eco-friendly and low operating cost. Low energy consumption.	Sensitivity to environmental conditions. Requires constant monitoring. Can collapse.
Biopercolator	Conversion to sulfuric acid or elemental sulfur through aerobic biological oxidation in a three-phase system.	Packed-bed bioreactor.	Improves biofilter efficiency by maintaining operating conditions.	Complexity in design and operation. Requires liquid effluent management. Can collapse.

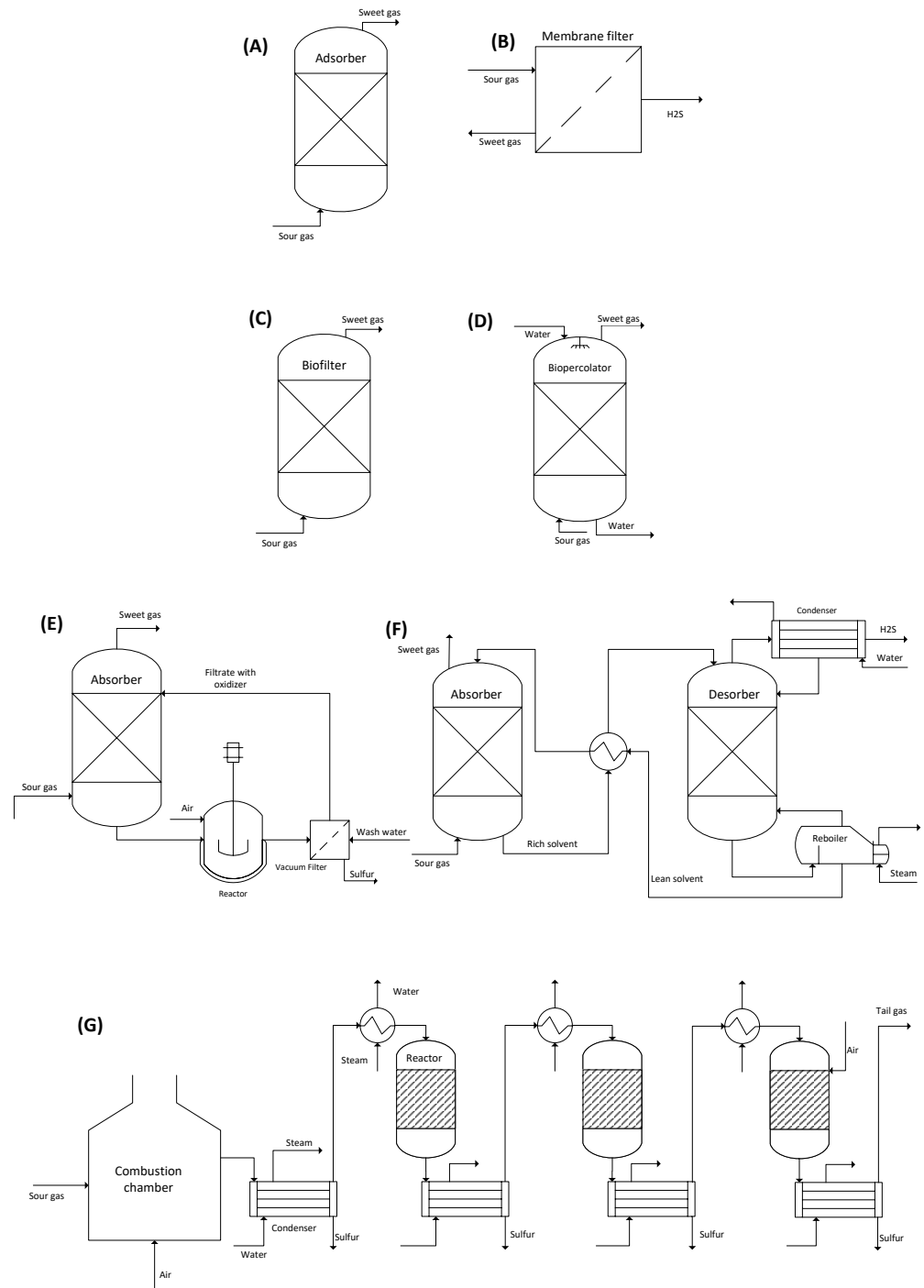


Figure 3. Summary of H_2S elimination processes. (A) Adsorption [285], (B) membrane filtration [286], (C) biofiltration [287], (D) biopercolation [288], (E) liquid oxidation [289], (F) absorption [290], and (G) Claus Process [291].

12. Conclusions

The multifaceted nature of the generation and emission of malodorous gases from aquatic environments, driven by microbial metabolism, nutrient imbalances, and anthropogenic stressors, demands a holistic approach to the system level for effective management. This review has synthesized the current understanding across the following key domains: fundamental microbial and biochemical mechanisms underlying H_2S , NH_3 , and VSC for-

mation; the influence of environmental parameters such as dissolved oxygen, organic load, and contaminant gradients; and the diverse range of advanced detection and mitigation techniques now available.

The interplay between regulatory oversight, scientific innovation, and community participation emerges as a crucial factor in mitigating these challenges. International standards, together with regionally adapted frameworks (exemplified by Colombian regulations), provide critical guidance for monitoring, compliance, and enforcement. However, significant improvements in odor control will require not only adherence to these standards but also adaptive management strategies that integrate continuous environmental sensing, predictive modeling, and targeted interventions.

On the detection front, while sophisticated laboratory instruments (e.g., GC-MS, GC-SCD, CRDS, QEPAS) have set new benchmarks for sensitivity and speciation, practical field applications increasingly favor portable, cost-effective, and real-time systems. Advances in electrochemical sensors, smartphone-based colorimetry, and optical spectroscopy pave the way for decentralized monitoring, enabling rapid response and adaptive odor management strategies.

Mitigation and control methods, ranging from classical physicochemical absorption and adsorption technologies to biologically driven treatments (biofilters, percolator biofilters) and chemical conversion processes, underscore the importance of context-specific solutions. Each strategy offers unique advantages and limitations that influence the selection criteria considering operational complexity, lifecycle costs, environmental impact, and scalability. Emerging hybrid technologies and integrated bioreactor designs promise to improve reliability, sustainability, and energy efficiency.

The case of the Bogotá River exemplifies the persistence and complexity of malodor problems in urbanized watersheds, where cumulative stresses from untreated wastewater, agricultural runoff, and industrial effluents converge. To restore water quality and alleviate the community's discomfort associated with odors, interventions must address root causes, such as infrastructure deficits, nutrient management, and stakeholder awareness, rather than relying solely on end-of-pipe treatments.

Moving forward, interdisciplinary research must bridge microbial ecology, environmental engineering, analytical chemistry, and policy studies. Innovations in biomolecular techniques, sensor miniaturization, and computational modeling will deepen our mechanistic understanding and improve our predictive capabilities. In parallel, the building of capacity among environmental regulators and stakeholders, combined with transparent data sharing and inclusive decision-making, can accelerate the transition to ecologically resilient, odor-free aquatic systems.

Ultimately, tackling malodorous gases requires a convergence of technological ingenuity, sound governance, and community involvement. By strengthening these links, we can forge more effective strategies to preserve environmental integrity, safeguard public health, and ensure the sustainable use of precious aquatic resources for future generations.

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